S1 File. Supporting information

DAPPER: Differential-Attention, Path-Personalised Federated Inference of Drone Flight Phase from Encrypted Cellular Command-and-Control Traffic

S1.1 Flight-phase labelling rules

Labels are assigned to every 100 ms bin from the drone's own state log (5 Hz: armed flag, flight mode, active mission command, altitude, NED velocity, manual set-point and stick-override flags). The rules are applied in the order listed; a later rule overrides an earlier one. v_h is the horizontal speed, v_z the vertical speed (positive upwards), h the altitude above the launch point.

Multirotor (ArduCopter).

1. default: *Cruise*;
2. $v_h < 0.7\,\mathrm{m/s}$: *Hover*;
3. $v_z < -0.4\,\mathrm{m/s}$ and $v_h < 1.5\,\mathrm{m/s}$: *Descent*; $v_z > 0.4\,\mathrm{m/s}$ and $v_h < 1.5\,\mathrm{m/s}$: *Ascent*;
4. mode CIRCLE, or mode AUTO with the active command LOITER_TURNS, and $v_h \geq 0.7\,\mathrm{m/s}$: *Orbit*;
5. a manual velocity set-point stream or an RC stick override is active: *Tele-operation*;
6. disarmed, or $h < 0.5\,\mathrm{m}$ with $v_h < 0.3\,\mathrm{m/s}$ and $|v_z| < 0.3\,\mathrm{m/s}$: *Ground*.

VTOL fixed-wing (ArduPlane QuadPlane).

1. default: *Cruise*;
2. mode LOITER or CIRCLE, or mode AUTO with command LOITER_UNLIM / LOITER_TURNS / LOITER_TIME / LOITER_TO_ALT: *Loiter*;
3. hover-borne flight: $v_h < 7\,\mathrm{m/s}$, or mode QHOVER / QLOITER / QLAND / QRTL, or mode AUTO with command VTOL_TAKEOFF / VTOL_LAND; of these, QLAND / QRTL, VTOL_LAND, RTL with $v_h < 7\,\mathrm{m/s}$, or $v_z < -0.3\,\mathrm{m/s}$ give *VTOL descent*, the rest *VTOL climb*;
4. an RC stick override is active: *Tele-operation*;
5. disarmed, or $h < 1\,\mathrm{m}$ with $v_h < 0.5\,\mathrm{m/s}$ and $|v_z| < 0.3\,\mathrm{m/s}$: *Ground*.

Operational phases group the fine labels as Ground; Stationary = {Hover, Orbit, Loiter}; Transit = {Ascent, Cruise, Descent, VTOL climb, VTOL descent}; Tele-operation. **Operational state** merges Stationary and Transit into Autonomous flight.

S1.2 Cell-site link and cross-traffic configuration

Each cell site is a router whose drone-facing links (uplink and downlink) and backhaul link to the core are shaped with Linux `netem` (delay, jitter, loss) and a token-bucket rate limit. Cross-traffic is generated by one emulated user-equipment node per site that alternates on/off sessions (exponential on and off durations with the listed means) at a rate drawn uniformly from the listed range, sharing the site's backhaul.

Site	Type	Uplink (delay ± jitter, loss, rate)	Downlink	Backhaul
A1	urban 5G	6(2) ms, 0.05%, 50	6(2) ms, 0.05%, 200	3.0(5) ms, 200
A2	suburban LTE	22(6) ms, 0.3%, 20	22(6) ms, 0.3%, 60	6(1) ms, 50
A3	rural LTE	40(15) ms, 1%, 5	40(15) ms, 1%, 15	12(2) ms, 20
A4	congested LTE	28(10) ms, 0.5%, 10	28(10) ms, 0.5%, 30	8(2) ms, 15
B1	5G SA	5.0(15) ms, 0.05%, 50	5.0(15) ms, 0.05%, 200	3.0(5) ms, 200
B2	LTE	25(8) ms, 0.4%, 20	25(8) ms, 0.4%, 60	8(1) ms, 50
B3	LEO satellite	15(4) ms, 0.2%, 30	15(4) ms, 0.2%, 100	20–55 ms varying every 15 s ±3 ms, 0.3%, 100
B4	GEO satellite	20(5) ms, 0.3%, 20	20(5) ms, 0.3%, 60	270(5) ms, 0.2%, 10

Rates are in Mbit/s. Telemetry stream profiles differ per site (high at A1/B1, standard at A2/A4/B2, low at A3/B3/B4), the companion forwarder batches frames for 0–100 ms depending on the site, and tele-operation set-points are sent at 10 Hz (A1, A4, B1) or 5 Hz (other sites).

S1.3 Mission types and parameter ranges

Each drone flies an endless sequence of missions drawn from its site's mix, with an idle period of 20–90 s between missions and a post-landing flight-log upload. Parameters are drawn uniformly per sortie.

Multirotor. *Delivery*: altitude 25–60 m, speed 6–12 m/s, distance 300–1100 m, random bearing, hover 5–25 s, descent to 6–10 m for a 5–15 s drop, return. *Survey*: altitude 30–70 m, speed 5–10 m/s, 4–8 legs of 150–380 m spaced 25–60 m, offset 60–250 m, camera trigger every 15–30 m. *Inspection*: altitude 20–50 m, distance 150–600 m, 2–4 orbit turns of radius 15–50 m, hold 15–45 s. *Tele-operation*: altitude 15–40 m, 60–180 s of manual velocity set-points at the site's rate.

VTOL. *Survey*: altitude 40–80 m, airspeed 16–22 m/s, 4–7 legs of 400–900 m spaced 80–150 m, offset 200–500 m. *Patrol*: altitude 50–100 m, airspeed 17–23 m/s, 3–5 legs of 800–2000 m. *Inspection*: altitude 40–80 m, airspeed 16–22 m/s, distance 500–1500 m, 2–5 orbit turns of radius 60–120 m. *Tele-operation*: altitude 50–90 m, airspeed 16–22 m/s, 60–150 s of manual input at 300–600 m.

Mission mixes (probabilities): A1 delivery .20, tele-operation .35, survey .15, inspection .30; A2 survey .25, delivery .15, inspection .30, tele-operation .30; A3 inspection .35, survey .20, delivery .15, tele-operation .30; A4 delivery .15, tele-operation .35, survey .20, inspection .30; B1 tele-operation .35, patrol .20, survey .15, inspection .30; B2 survey .20, patrol .20, inspection .30, tele-operation .30; B3 patrol .20, inspection .35, survey .15, tele-operation .30; B4 inspection .35, survey .20, patrol .15, tele-operation .30.

S1.4 Window counts per site, split and fine phase

Multirotor fleet (A).

Site	Split	Windows	Ground	Ascent	Hover	Cruise	Orbit	Teleop	Descent
A1	train	2187	520	95	174	626	418	73	281
A1	val	539	79	33	18	288	37	0	84
A1	test	217	32	7	18	62	77	0	21
A2	train	1862	554	73	66	452	193	325	199
A2	val	517	68	22	44	39	211	59	74
A2	test	391	115	29	15	156	0	0	76
A3	train	1611	639	68	27	238	135	300	204
A3	val	328	114	23	7	80	0	50	54
A3	test	404	175	5	11	119	0	64	30
A4	train	1898	521	89	74	514	138	338	224
A4	val	134	122	0	0	0	0	0	12
A4	test	492	138	15	18	177	92	0	52

VTOL fleet (B).

Site	Split	Windows	Ground	VTOL climb	VTOL descent	Cruise	Loiter	Teleop
B1	train	2040	638	123	260	541	286	192
B1	val	824	140	50	79	491	0	64
B1	test	324	61	14	18	231	0	0
B2	train	1898	582	97	88	912	36	183
B2	val	521	125	46	43	305	2	0
B2	test	718	230	46	37	255	0	150
B3	train	2423	756	126	224	493	681	143
B3	val	397	194	21	38	26	0	118
B3	test	462	123	25	41	22	251	0
B4	train	1670	1398	23	38	147	64	0
B4	val	388	276	8	23	11	0	70
B4	test	373	287	14	18	11	0	43

S1.5 Public PiNcH captures

The public dataset of Sciancalepore et al. (Computer Networks 168:107044, 2020; `https://cri-lab.net/pinch`) was converted with `code/public_pinch.py`. Steady/moving subset: 174,679 frames, 1202 s, 12,020

bins (6,013 steady, 6,007 moving), 30 s blocks, 12.8 s windows: 239 training, 48 validation and 47 test windows. Seven-movement subset: 66,317 frames, 430 s, 4,301 bins, 15 s blocks, 6.4 s windows, the last block of every movement held out (21 test windows).